Electricity

The big idea

Electricity powers almost everything around us. This chapter explains what makes charge flow, how resistance controls that flow, and how electrical energy turns into heat.

Current and potential difference

ELECTRIC CURRENT

$I = Q / t$ (unit ampere, A; 1 A = 1 C/s)

Charge flows only in a **closed circuit**. Conventional current flows from + to –, opposite to the flow of electrons. The charge of one electron is 1.6×10^{-19} C. An **ammeter** measures current and is connected in **series**.

POTENTIAL DIFFERENCE

$V = W / Q$ (unit volt, V; 1 V = 1 J/C)

A cell maintains a potential difference across the circuit. A **voltmeter** is connected in **parallel**.

Ohm's law and resistance

OHM'S LAW

At constant temperature, the current through a conductor is proportional to the potential difference across it: **$V = IR$**. R is the **resistance**, measured in

ohms (Ω). The V–I graph is a straight line through the origin.

Resistance depends on the conductor:

RESISTIVITY

$R = \rho l / A$

R increases with length l and decreases with area of cross-section A. ρ (resistivity, Ω m) depends on the material and temperature.

Metals have low ρ (copper, aluminium: used for wires). Alloys like **nichrome** have high ρ and don't oxidise easily (used in heater coils). Glass and rubber are insulators.

Combining resistors

SERIES AND PARALLEL

Series: $R = R_1 + R_2 + R_3$ (same current through each; voltages add)

Parallel: $1/R = 1/R_1 + 1/R_2 + 1/R_3$ (same voltage across each; currents add)

RESISTORS OF 2 Ω, 3 Ω AND 6 Ω ARE CONNECTED IN PARALLEL TO A 12 V BATTERY

$1/R = 1/2 + 1/3 + 1/6 = 1 \Rightarrow$ **R = 1 Ω**. Current from the battery $I = 12/1 =$ **12 A**.

Household appliances are connected in **parallel** so each gets the full voltage and can be switched on or off independently. The parallel resistance is **less** than the smallest individual resistance.

Heating effect and power

JOULE'S LAW OF HEATING AND POWER

$$H = I^2Rt$$

$$P = VI = I^2R = V^2/R \text{ (unit watt, W)}$$

$$\text{Electrical energy (commercial unit)} = 1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$$

The heating effect is used in electric irons, toasters, kettles, bulbs (tungsten filament, filled with nitrogen or argon) and **fuses** (a thin wire with a low melting point that melts and breaks the circuit when too much current flows).

A 100 W BULB RUNS 5 HOURS A DAY. HOW MANY UNITS DOES IT USE IN 30 DAYS?

Energy = 0.1 kW × 5 h × 30 = **15 kWh (units)**.

AMMETER AND VOLTMETER POSITIONS

Ammeter in **series** (it must carry the current). Voltmeter in **parallel** (it compares two points).

Quick recap

- $I = Q/t$, $V = W/Q$, $V = IR$.
- $R = \rho l/A$; series resistances add, parallel reciprocals add.
- $H = I^2Rt$; $P = VI$; 1 kWh = 3.6 MJ.

Practice questions

1. The unit of resistivity is
   (a) ohm (b) ohm metre (c) ohm per metre (d) volt per ampere
2. When the length of a wire is doubled (same material and thickness), its resistance
   (a) halves (b) doubles (c) stays the same (d) becomes four times
3. Two 4 Ω resistors in parallel have an equivalent resistance of
   (a) 2 Ω (b) 4 Ω (c) 8 Ω (d) 16 Ω
4. An electric heater draws 5 A at 220 V. Its power is
   (a) 44 W (b) 1100 W (c) 225 W (d) 1.1 W
5. A fuse works on the
   (a) magnetic effect of current (b) heating effect of current (c) chemical effect of current (d) photoelectric effect
6. A voltmeter is connected in
   (a) series (b) parallel (c) either way (d) neither

Answers: 1. b 2. b 3. a 4. b 5. b 6. b